

Fault-Tolerant Decentralized Coordination of Heterogeneous Robot Swarms for Warehouse Automation

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Problem Statement

- Modern robot fleets depend on centralized controllers that allocate tasks from a single computational unit(traffic controller).
- As fleet grows, this creates scalability limits and a single point of failure.
- Most existing decentralized approaches ignore heterogeneity constraints like speed, load capacity and battery leading to poor task-to-robot matching and insufficient movement. Constant peer-to-peer interaction also adds network traffic that worsens with fleet size.

MOTIVATION :

Develop a decentralized, capability-aware, and fault-tolerant coordination framework that enables efficient, scalable, and reliable warehouse automation.

1 SINGLE POINT OF FAILURE

3 COUPLED PROBLEMS

50+ HETEROGENEOUS ROBOTS

Literature Review

15

journal papers
reviewed

Task Allocation & Scheduling

- Mayya et al. (2021)
- Kalempa et al. (2021)
- Dai et al. (2025)
- Arbanas et al. (2024)
- Bagchi et al. (2024)
- Li et al. (2020)
- Notomista et al. (2022)

Decentralized Coordination & Communication

- Doostmohammadian et al. (2025)
- De Ryck et al. (2022)
- Lee et al. (2023)
- Ye et al. (2021)
- Sun & Shao (2022)

Fault Tolerance, Navigation & Resilience

- Neville et al. (2023)
- Chen et al. (2021)
- Miloradović et al. (2022)

Sl.No.	Author(s) & Journal Title	Year	Methodology/ Approach	Limitations & Future Scope
1	Mayya, S., D'Antonio, D.S., Saldaña, D., Kumar, V. — Resilient Task Allocation in Heterogeneous Multi-Robot Systems	2021	Optimization-based task allocation that tracks task performance deviation under disturbances and quantifies degradation of individual robot capabilities (e.g., sensing, mobility); reallocates robots to tasks by minimally relaxing resource constraints to preserve required aggregate capability per task	Assumes capability degradation can be measured/estimated in real time; centralized optimization formulation, not fully decentralized; does not address communication loss or complete robot failure (dropout), only capability degradation — future work could extend to decentralized solvers and abrupt failures, which is a direct gap this project addresses
2	Kalempa, V.C., Piardi, L., Limeira, M., de Oliveira, A.S. — Multi-Robot Preemptive Task Scheduling with Fault Recovery: A Novel Approach to Automatic Logistics of Smart Factories	2021	Multi-Robot Preemptive Task Scheduling with Fault Recovery (MRPF): introduces priority-based preemptive scheduling with task dependencies, managing process/task interaction at each event to minimize idleness and recover automatically from robot faults	Tested only on a small robot team in a constrained testbed (limited scalability evidence); does not explicitly model heterogeneous robot capabilities (speed/load/battery) in the bidding process — a direct gap this project's capability-aware allocation addresses
3	Ye, F., Chen, J., Sun, Q., Tian, Y., Jiang, T. — Decentralized Task Allocation for Heterogeneous Multi-UAV System with Task Coupling Constraints	2021	Extends CBBA into CBBA-TCC (Task Coupling Constraints) — a two-stage inner/outer consensus algorithm where each agent builds a "Can-do list" of tasks it's capable of performing without violating precedence constraints, then bids only within that list, enabling decentralized allocation for genuinely heterogeneous agents	It's essentially CBBA extended for heterogeneity, however it targets UAV swarms (aerial, not ground/warehouse), assumes static task coupling graphs, and doesn't address fault tolerance or robot failures
4	Chen, L., Zhao, Y., Zhao, H., Zheng, B. — Non-Communication Decentralized Multi-Robot Collision Avoidance in Grid Map Workspace with Double Deep Q-Network	2021	Decentralized, communication-free collision avoidance policy learned via DDQN that lets each robot avoid collisions using only local Lidar-like sensing (no inter-robot messaging), explicitly designed for large-scale grid-map workspaces resembling automated warehouses	Communication-free, so it can't leverage inter-robot coordination for better solutions in dense scenes; ignores heterogeneous robot capabilities and task allocation entirely; trained on a fixed grid layout, so generalization to other warehouse configurations is untested.
5	Li, Z., Barenji, A.V., Jiang, J., Zhong, R.Y., Xu, G. — A Mechanism for Scheduling Multi-Robot Intelligent Warehouse System Facing Dynamic Demand	2020	Proposes a scheduling mechanism for multi-robot and task-allocation problems in automated storage/retrieval warehouses that responds to simultaneous, dynamically arriving customer orders, aiming to avoid	The scheduling mechanism is not explicitly decentralized or agent-distributed — coordination logic appears to sit closer to a centralized/semi-centralized scheduler; does not model heterogeneous robot capabilities (speed, load capacity, battery) in the allocation decision; fault tolerance and robot failure recovery are not addressed; scalability beyond the tested order-volume scenarios is not evaluated.

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6	M. Doostmohammadian, A. Aghasi, M. Pirani, E. Nekouei, H. Zarrabi, R. Keypour, A. I. Rikos & K. H. Johansson – "Survey of Distributed Algorithms for Resource Allocation over Multi-Agent Systems"	2025	It provides a mathematical formulation of distributed resource allocation and gives an overview of primal-based, dual-based, and ADMM-based distributed approaches. The paper further reviews issues like feasibility, convergence, network reliability, and applicability in areas such as smart grids and scheduling.	The paper basically gives only a description of the methods and does not validate the novel methods on robotic datasets. In the paper some topics like non-linear dynamics, delay in communication and large networks are highlighted.
7	M. De Ryck, D. Pissoort, T. Holvoet & E. Demeester – "Decentral Task Allocation for Industrial AGV-Systems with Routing Constraints"	2022	In the paper a multi-agent system is introduced so that tasks are allocated via auction and routing is delegated to avoid congestion and achieve better scalability for warehouse logistics.	In the paper it is assumed that AGVs are uniform and communications are reliable. However, robots have different capabilities, and it is possible that a robot can fail. Future work is planned to use the method with heterogeneous agents, dynamic environment and fault tolerant coordination.
8	W. Dai, U. Rai, J. Chiun, C. Yuhong & G. Sartoretti – "Heterogeneous Multi-robot Task Allocation and Scheduling via Reinforcement Learning"	2025	Proposes a framework based on reinforcement learning for decentralized heterogeneous allocation and scheduling with the use of attention mechanisms that can model short-term cooperation and long-term scheduling problems. Also we introduce the strategy of constrained flashforward to help with exploration.	The process is decentralized in terms of its implementation but centralized in terms of the learning process. It is assumed that communication is reliable; robot failure/recovery issues are not considered.
9	B. Arbanas, T. Petrović, M. Orsag, J. R. Martínez-de-Dios & S. Bogdan – "Distributed Allocation and Scheduling of Tasks With Cross-Schedule Dependencies for Heterogeneous Multi-Robot Teams"	2024	The issue is defined as a distributed vehicle routing/task scheduling issue, and a distributed evolutionary optimization algorithm (CBM-pop) is presented, which allows decentralized negotiation along with addressing precedence and timing constraints between different robots.	The model assumes reliable communication and does not include fault detection or recovery. Computational complexity increases with highly dynamic environments. Future work includes real-world deployment, improved scalability, communication robustness, and online replanning under failures.
10	M. J. Bagchi, S. B. Nair & P. K. Das – "On a Dynamic and Decentralized Energy-Aware Technique for Multi-Robot Task Allocation"	2024	Introduces a decentralized auction-based MRTA algorithm that incorporates battery level into bid computation. The algorithm makes use of mobile agents for resolving conflicts, adaptive energy thresholds for charging the robot, and mechanisms for dynamically handling situations like task arrivals and execution delays.	The core focus of the algorithm is energy efficiency, not heterogeneous capability modeling. There isn't any fault detection or recovery. Future work includes heterogeneous robot support, and learning-based task allocation, and fault-tolerant decentralized coordination.

Sl.No.	Author(s) & Journal Title	Year	Methodology/ Approach	Limitations & Future Scope
11	Lee, H.-Y., Zhou, P., Zhang, B., Qiu, L., Fan, B., Duan, A., Tang, J., Lam, T.L., Navarro-Alarcon, D. — A Distributed Dynamic Framework to Allocate Collaborative Tasks Based on Capability Matching in Heterogeneous Multirobot Systems	2023	Completely decentralized system where the capabilities of robots are matched to task requirements in real-time, thus forming an ad-hoc coalition in order to work on collective tasks, without any central control; robots bargain locally to redistribute the sub-tasks in case of capability mismatches.	Assumption of truthful reported profiles of capabilities of robots, as well as effective communications between robots locally; does not model the issue of sudden failure and dropout in either case.
12	Neville, G., Chernova, S., Ravichandar, H. — D-ITAGS: A Dynamic Interleaved Approach to Resilient Task Allocation, Scheduling, and Motion Planning	2023	Brings together task allocation, scheduling, and path planning into one optimization cycle; dynamically re-solves in an incremental manner upon disruption with the goal of maintaining schedule feasibility and robustness without doing a full replan	Centralized/interleaved solver architecture (not decentralised) Evaluated mainly in simulation with idealised trait models Does not consider communication-constrained or fully distributed warehouse deployments
13	Miloradović, B., Ćurić, B., Ekström, M., Papadopoulos, A.V. — GMP: A Genetic Mission Planner for Heterogeneous Multirobot System Applications	2022	Genetic-algorithm-based mission planner that searches task-to-robot assignments and orderings to satisfy heterogeneous skill and precedence constraints, aiming for near-optimal mission plans that can be regenerated on-line	Evolutionary search has non-deterministic runtime, thus constraining the possibility for real-time replanning after faults; it is not explicitly designed for fault-tolerance or decentralisation - planning still remains a single global optimisation, a direct gap this project addresses
14	Sun, Y., Shao, Y. — Research on Data Security Communication Scheme of Heterogeneous Swarm Robotics System in Emergency Scenarios	2022	Decentralized Master-Slave Blockchain with Delegated Proof of Stake Consensus for maintaining data consistency and fault tolerance from node failures in a heterogeneous swarm without a fixed central node	Only validated through simulations of a network inside a container; it concentrates on the fault tolerance of the data/communication layer, not on task assignment or motion planning – does not consider task reallocation based on capability awareness.
15	Notomista, G., Mayya, S., Emam, Y., Kroninger, C., Bohannon, A., Hutchinson, S., Egerstedt, M. — A Resilient and Energy-Aware Task Allocation Framework for Heterogeneous Multi-Robot Systems	2022	Constraint-based optimization scheme on the internet where robot "characteristics" are separated from "abilities" in tasks, enabling the reallocation of characteristics to cope with their failures, while optimizing energy usage.	The framework is centralized/semi-centralized in terms of optimization; concentrates on energy survivability instead of warehouse efficiency; little testing done at the swarm level (the robot-based experiments involve small swarms).

Research Gap

Heterogeneity and Fault Tolerance are Treated Separately

Existing approaches either optimize task allocation based on robot capabilities or focus on recovering from robot failures. Very few frameworks integrate both, resulting in inefficient task reassignment when failures occur.

Dependence on Centralized Coordination

Many capability-aware systems still rely on centralized or semi-centralized planners, creating computational bottlenecks and a single point of failure, limiting scalability in large warehouse robot fleets.

Limited Fault Handling in Real-World Scenarios

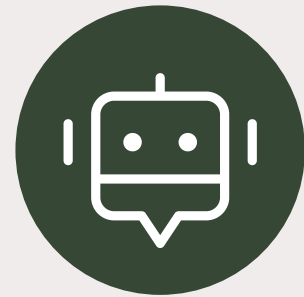
Most studies only consider capability degradation or simple robot failures. Practical issues such as communication loss, battery depletion, sudden robot crashes, and dynamic task reallocation remain largely unaddressed.

High Communication Overhead in Decentralized Systems

Current decentralized coordination often requires continuous communication between robots, leading to excessive network traffic and reduced scalability as swarm size increases.

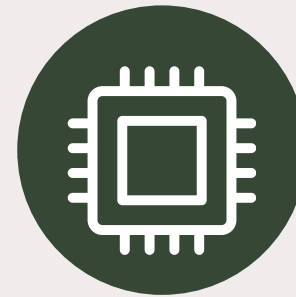
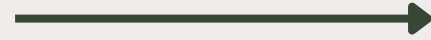
Together: No existing framework simultaneously provides capability-aware task allocation, fully decentralized coordination, event-driven communication, and robust fault recovery for heterogeneous warehouse robot swarms.

Proposed Solution



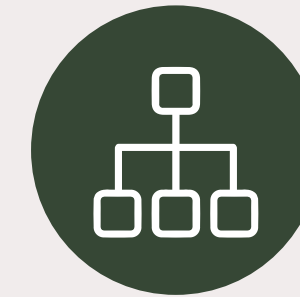
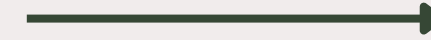
Heterogeneous Fleet

Robots modeled with distinct speed, load, battery & task-suitability profiles.



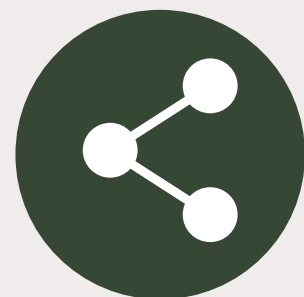
Capability -Aware Bid

Each robot bids using distance, battery, workload, congestion and capability match.



Event -Triggered Consensus

Robots exchange info only on triggering events, reducing communication overhead.



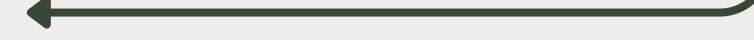
Local Reallocation

On fault detection, neighbours reassign the failed robot's tasks with no central authority



Heartbeat Monitoring

Neighbouring robots continuously watch for faulty or unresponsive peers.



Research Objectives

Objective 01

Model a heterogeneous warehouse robot swarm with robots differing in speed, load capacity, battery usage, and task suitability.

Objective 02

Design a decentralized bidding mechanism that assigns tasks based on distance, battery level, workload, congestion, and robot-task capability match — learned via reinforcement learning rather than hand-coded rules.

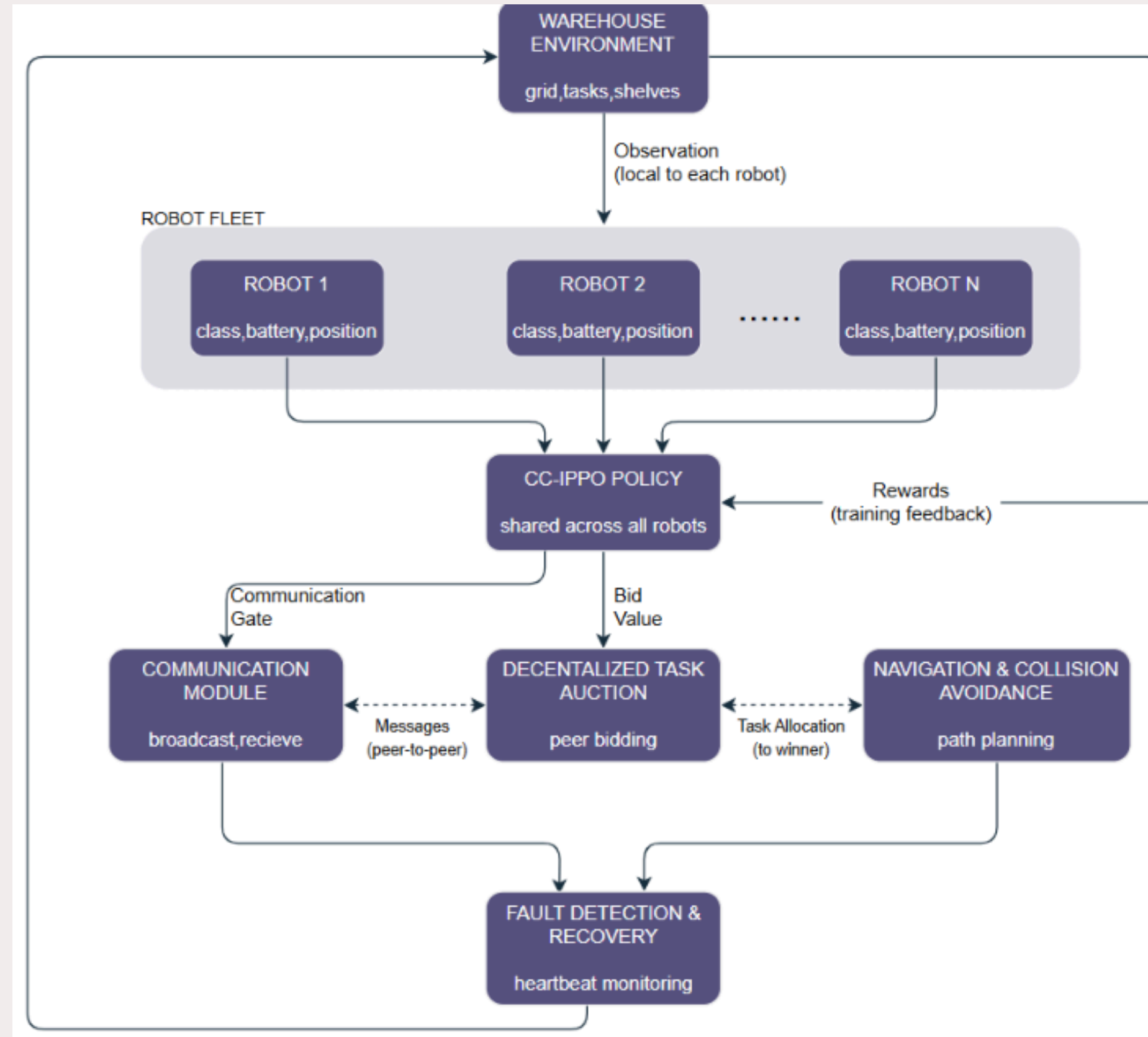
Objective 03

Implement event-triggered communication, local consensus, and decentralized collision avoidance for efficient robot coordination with minimal network overhead.

Objective 04

Integrate heartbeat-based fault detection and local task reallocation to recover from robot failures without a central controller, and evaluate scalability from approximately 4 to 50+ agents.

Overall System Architecture



Module 1 – Warehouse Environment & Heterogeneity Model

Purpose : Simulate the warehouse and provide a heterogeneous robotfleet, instead of treating all robots identical

Capability Class

Each robot is light-duty, heavy-load or balanced

Battery Model

Class-specific Battery consumption rates

Speed Differentiation

Light-duty robots move one extra grid/time-unit

Task Typing

Tasks assigned by self weight and distance

Mismatch Penalty

Penallizes delivering an unsuited shelf type

Output : RWARE's base 71-dim observation is expanded to 80 dim(+3 classes, +1 battery, +5 task) – fed directly into the CC-IPPO policy

Module 2 – CC-IPPO Policy (Capability – Conditioned Independent PPO)

Purpose : Give every robot a decision-making capability that determines how strongly to bid for a nearby task and whether to broadcast a message — learned through trial and error rather than hand-coded rules.

Shared Network

every robot runs an identical copy of the same policy (Independent PPO); no separate network per robot class needed

Input

the 80-value observation vector from the environment

Two Output Heads

Bid Value (0 to 1, task suitability) and Communication Gate (broadcast or not)

Reward Signal

combines task throughput, capability-mismatch penalty, collision penalty, communication-cost penalty, and fault-recovery bonus

Decentralized Training

each robot collects and learns from its own experience independently, with no central learner

Output : After training, we validate capability-awareness directly — a heavy-load robot's bid for a heavy-shelf task should exceed its bid for an urgent-light task

Module 3 & 4- Learned Communication and Decentralized Auction

3 Learned Communication

- Communication-gate output from the CC-IPPO policy decides, per robot per step, whether the benefit of broadcasting outweighs its cost
- If enabled, the robot broadcasts a short message (state, bid intent, task-relevant data) to neighbors within a fixed communication radius
- Channel models a 5% message-loss rate, which Fault Detection (Module 6) is specifically designed to tolerate
- Communication-cost penalty in the reward function drives the policy to communicate only when it actually changes an outcome

4 Decentralized Task Auction

- When a task appears, all nearby robots compute a bid using their CC-IPPO policy's bid-value head
- Bids are collected and the highest bidder wins — no central scheduler involved
- Ties broken by priority order: capability match first, then battery level, then workload
- Winning robot is routed to its PathPlanner (Module 5) for execution

Module 5 & 6 - Navigation & Fault Detection/ Recovery

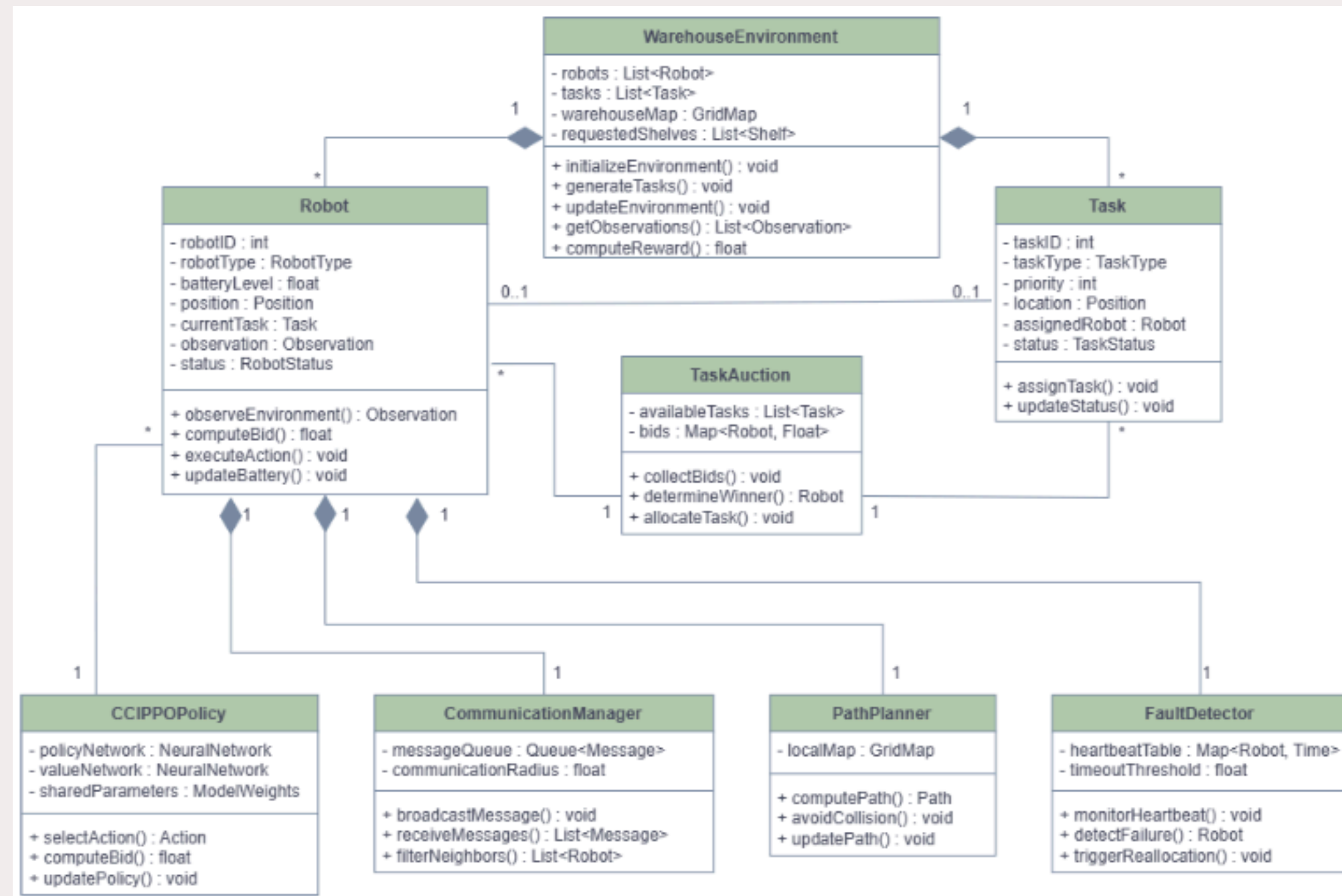
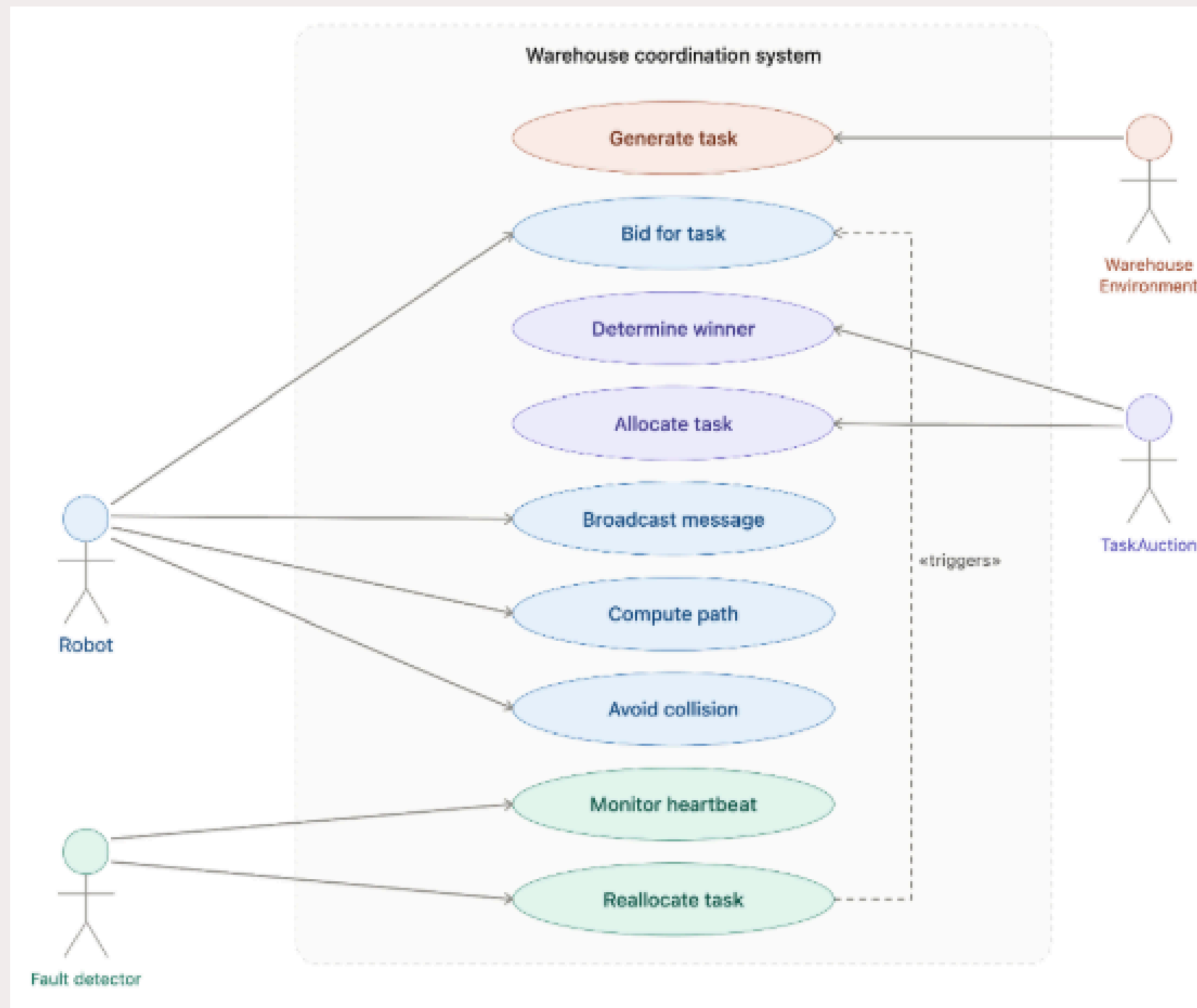
5 Navigation & Collision Avoidance

- A* path search from the current cell to the goal task, computed locally by each robot.
- Conflict priority order: loaded robots > urgent-task robots > low-battery robots, with a tie-break to prevent starvation.
- Heterogeneity interacts directly here — fast_light robots occasionally get an extra move; heavy_load robots occasionally skip one, requiring path recomputation.

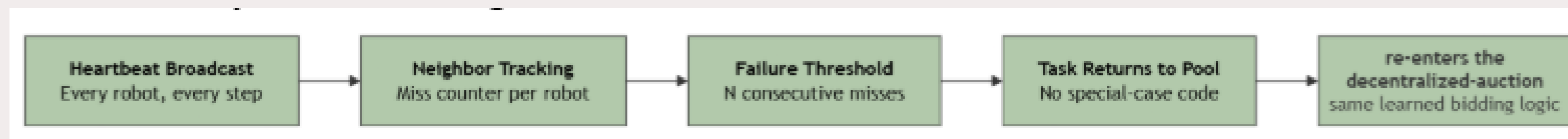
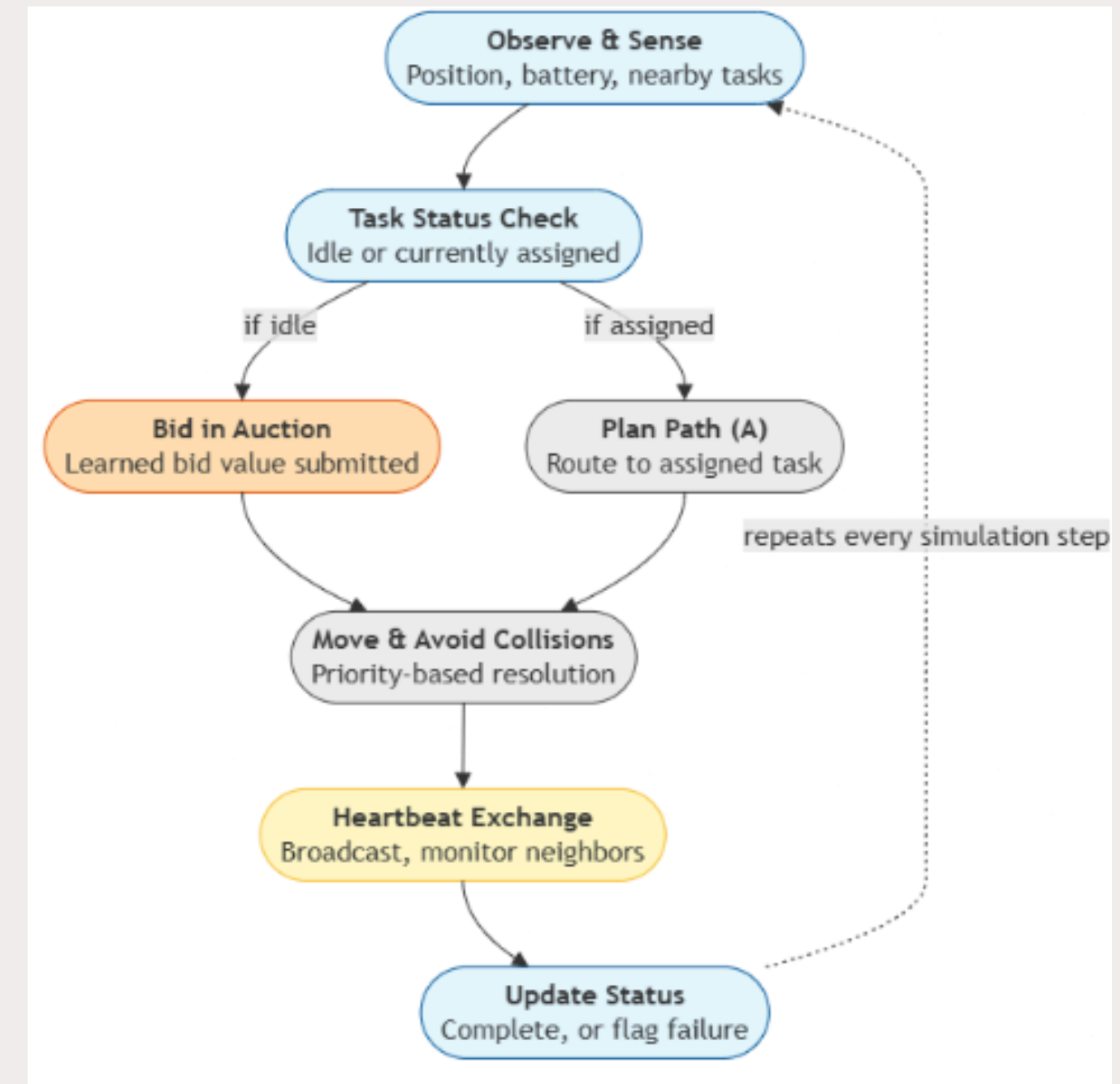
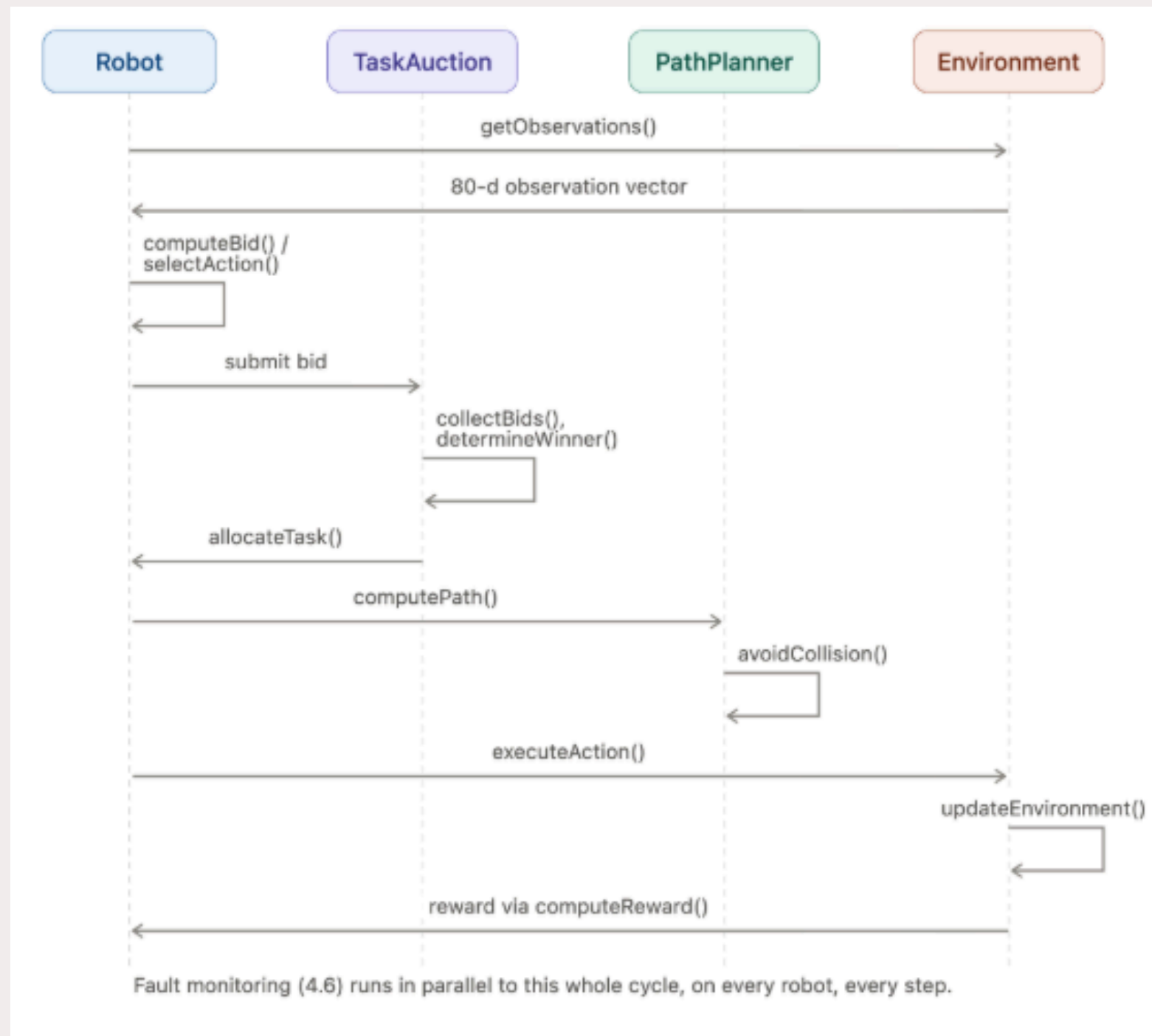
6 Fault Detection & Recovery

- Heartbeat mechanism: every robot signals “I am alive” each step to nearby robots.
- Failure threshold: 3 consecutive missed heartbeats (tolerates the 5% communication-loss rate) before a robot is marked faulty.
- Task recovery: the failed robot's task re-enters the pool and is re-won through the same running auction — no special-case recovery code required.
- Evaluated at induced fault rates of 0%, 10%, 20%, 30%, and 50%.

UML Diagrams - Use Case & Class Diagram



Sequence & Workflow Diagram



Performance Evaluation

Primary Performance

- Task Throughput — Total number of successful deliveries completed.
- Capability Match Rate — Percentage of deliveries handled by a robot with suitable capabilities.
- Capability Mismatch Rate — Percentage of deliveries performed by an unsuitable robot.

Robustness & Scalability

- Failure Degradation — Change in throughput and match rate as robots are progressively removed/fail.
- Per-Robot Throughput — Average deliveries achieved per robot as swarm size increases.
- Communication Ablation — Compare system performance with and without communication to measure the contribution of decentralized communication.

Evaluation Scenarios

- Capability Matching: Verify that robots are appropriately matched to tasks.
- Fault Tolerance: Evaluate performance under robot failures.
- Scalability: Evaluate performance across different swarm sizes.
- Communication Ablation: Determine the impact of peer-to-peer communication.

Success Criterion: The system should maintain effective task allocation and degrade gracefully under robot failures while preserving capability-aware task matching.

Implementation Status



References

- S. Mayya, D. S. D'Antonio, D. Saldaña, and V. Kumar, "Resilient Task Allocation in Heterogeneous Multi-Robot Systems," IEEE Robotics and Automation Letters, 2021.
- V. C. Kalempa, L. Piardi, M. Limeira, and A. S. de Oliveira, "Multi-Robot Preemptive Task Scheduling with Fault Recovery: A Novel Approach to Automatic Logistics of Smart Factories," Sensors, 2021.
- F. Ye, J. Chen, Q. Sun, Y. Tian, and T. Jiang, "Decentralized Task Allocation for Heterogeneous Multi-UAV System with Task Coupling Constraints," The Journal of Supercomputing, 2021.
- L. Chen, Y. Zhao, H. Zhao, and B. Zheng, "Non-Communication Decentralized Multi-Robot Collision Avoidance in Grid Map Workspace with Double Deep Q-Network," Sensors, 2021.
- Z. Li, A. V. Barenji, J. Jiang, R. Y. Zhong, and G. Xu, "A Mechanism for Scheduling Multi-Robot Intelligent Warehouse System Facing Dynamic Demand," Journal of Intelligent Manufacturing, 2020.
- M. Doostmohammadian, A. Aghasi, M. Pirani, E. Nekouei, H. Zarrabi, R. Keypour, A. I. Rikos, and K. H. Johansson, "Survey of Distributed Algorithms for Resource Allocation over Multi-Agent Systems," 2025.
- M. De Ryck, D. Pissoort, T. Holvoet, and E. Demeester, "Decentral Task Allocation for Industrial AGV-Systems with Routing Constraints," 2022.
- W. Dai, U. Rai, J. Chiun, C. Yuhong, and G. Sartoretti, "Heterogeneous Multi-robot Task Allocation and Scheduling via Reinforcement Learning," 2025.
- B. Arbanas, T. Petrović, M. Orsag, J. R. Martínez-de-Dios, and S. Bogdan, "Distributed Allocation and Scheduling of Tasks With Cross-Schedule Dependencies for Heterogeneous Multi-Robot Teams," 2024.
- M. J. Bagchi, S. B. Nair, and P. K. Das, "On a Dynamic and Decentralized Energy-Aware Technique for Multi-Robot Task Allocation," 2024.
- H.-Y. Lee, P. Zhou, B. Zhang, L. Qiu, B. Fan, A. Duan, J. Tang, T. L. Lam, and D. Navarro-Alarcon, "A Distributed Dynamic Framework to Allocate Collaborative Tasks Based on Capability Matching in Heterogeneous Multirobot Systems," 2023.
- G. Neville, S. Chernova, and H. Ravichandar, "D-ITAGS: A Dynamic Interleaved Approach to Resilient Task Allocation, Scheduling, and Motion Planning," 2023.
- B. Miloradović, B. Ćurić, M. Ekström, and A. V. Papadopoulos, "GMP: A Genetic Mission Planner for Heterogeneous Multirobot System Applications," 2022.
- Y. Sun and Y. Shao, "Research on Data Security Communication Scheme of Heterogeneous Swarm Robotics System in Emergency Scenarios," 2022.
- G. Notomista, S. Mayya, Y. Emam, C. Kroninger, A. Bohannon, S. Hutchinson, and M. Egerstedt, "A Resilient and Energy-Aware Task Allocation Framework for Heterogeneous Multi-Robot Systems," 2022.

Thank You